TECHNO
TECHNOPRENEURSHIP
2nd Semester, SY 2025 – 2026

GROUP ACTIVITY NO. 1
ENGINEER'S INNOVATION CANVAS

GENERAL INSTRUCTION: Use this template exactly as provided. Do not modify the paper size, margins, paragraph spacing, font type, font size, or any other formatting. Simply type your answers directly into the template. After completing the document, save it as a PDF using the following filename format:

`techno (wednesday) group_name – engineers innovation canvas.pdf`

Note: the filename must be in small letters only.

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In this activity, you will work in groups to think like computer engineers and technopreneurs. You will take a common household item and redesign it into a smart or automated device using technology. This activity will help you connect engineering ideas with real-world problems, innovation, and scalability. There is no right or wrong answer—focus on being creative, practical, and clear in explaining your solution.

- Choose ONE common household item.
- Convert it into a smart or automated device.
- Complete all sections of the Innovation Canvas.
- Be clear, realistic, and concise.
- Prepare to briefly explain your idea.

Answer the following questions: **Limit your answer within the space provided.**

HOUSEHOLD ITEM	**Circuit Breaker**

PROBLEM	What problem does this household item currently have? *Tips on how to answer:* ▪ *Identify a real, common problem* ▪ *Focus on safety, convenience, time, cost, or efficiency* ▪ *Avoid vague answers* ▪ *Example:* ✔ *Good answer: "People forget to turn it off."* ❌ *Weak answer: "It is not good."*
ANSWER	
In traditional circuit breakers, users cannot remotely monitor the power usage, faults, or energy waste. Traditional circuit breakers only cut power during overload or short circuit and do not provide real-time monitoring, data, alerts, and usage. Users are unaware of overheating, abnormal power consumption, or potential faults until damage, downtime, or fire risk occurs. Manual inspections are time-consuming, inconvenient, and unsafe in some situations.	

TARGET USER	Who will use the device?
	Tips on how to answer:
	▪ *Be specific*
	▪ *Mention who, where, or situation*
	▪ *Example:*
	  ▪ *Busy working adults*
	  ▪ *Families with children*
	  ▪ *Elderly people*

ANSWER	
● Homeowners in areas with unstable power grids ● Remote workers and hybrid employees ● Households with high energy consumption	

TECHNOLOGY SOLUTION	What technology will you use to solve the problem?
	Tips on how to answer:
	▪ *Focus on **engineering components***
	▪ *Technology must be the **core solution***
	▪ *Examples:*
	  ▪ *Sensors*
	  ▪ *Microcontrollers*
	  ▪ *Mobile apps*
	  ▪ *IoT systems*
	
	❗ *Do NOT list social media or marketing tools.*

ANSWER	

The solution will use IoT technology as the core system, combining electrical sensors to measure voltage, current, and temperature with a microcontroller that processes data in real time. When abnormal conditions are detected, the microcontroller automatically controls the circuit breaker to disconnect power. The system connects wirelessly to a mobile application, allowing users to monitor energy usage, receive alerts, and remotely control circuits, making the solution smart, automatic, and reliable.

INNOVATION	What makes your solution better than the traditional one? *Tips on how to answer:* *Answer at least ONE:* ▪ *Is it faster?* ▪ *Safer?* ▪ *Cheaper?* ▪ *Automatic?* ▪ *Smarter?* *Use phrases like:* ▪ *"Unlike the traditional version…"* ▪ *"This improves the current solution by…".*

ANSWER

Unlike the traditional circuit breaker, which only reacts after an overload occurs, the Smart Circuit Breaker continuously monitors electrical conditions in real time. This improves the current solution by being safer, smarter, and more automated, as it detects electrical issues early and allows remote control, monitoring, and resetting through a mobile app, providing better protection and convenience than traditional systems.

RISKS	What could go wrong with your idea? *List **at least two risks**:* ▪ 🔧 *Technical (device failure, sensors, power)* ▪ 📉 *User (people may not use it)* ▪ 💰 *Cost (expensive to build or maintain)* *Be honest and realistic.*

ANSWER

One possible risk is technical issues, such as sensor inaccuracies or device failure, which could lead to false alarms or missed electrical problems. The system may also depend on stable power and internet connectivity, and losing these could limit remote monitoring features. Another risk is cost, as IoT sensors and controllers can be expensive to install and maintain. Additionally, some users may be hesitant to adopt new technology or may not fully utilize the mobile app, thereby reducing the system's overall effectiveness.

SCALABILITY	How can this idea grow or be used by more people? *Tips on how to answer:* ***Think beyond one house:*** ▪ ***Can it be used in other homes?*** ▪ ***Apartments?*** ▪ ***Schools?*** ▪ ***Offices?***

ANSWER

This smart breaker is highly applicable in residential households, giving homeowners detailed data on energy-hungry appliances and better fire safety. Apart from that, it is vital for multi-tenant buildings like apartments and condos, allowing managers to reset switches remotely and handle power in vacant units. Furthermore, it scales easily to commercial and institutional settings, such as schools, office towers, and retail centers, where it replaces manual work with automated schedules to save money. Ultimately, by bringing IoT control to everything from a single-family home to large industrial factories, the electrical panel becomes a proactive tool that predicts equipment failures, cuts costs, and protects lives on a massive scale.

SMART IoT-BASED CIRCUIT BREAKER

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